

Section 1 (Verification): Last 10 Rows of Model Dataset

Obs	Month	CPI
60	DEC2016	106.2
61	JAN2017	106.2
62	FEB2017	106.2
63	MAR2017	106.2
64	APR2017	106.2
65	MAY2017	106.3
66	JUN2017	106.3
67	JUL2017	106.2
68	AUG2017	106.3
69	SEP2017	106.4
70	OCT2017	.

CPI should show missing (.) for all rows after 30SEP2017

Section 2: Unobserved Components Model (UCM)

The UCM Procedure

Input Data Set	
Name	WORK.MODEL
Time ID Variable	Month

Estimation Span Summary									
Variable	Type	First Obs	Last Obs	NObs	NMiss	Min	Max	Mean	Standard Deviation
CPI	Dependent	JAN2012	MAR2017	63	0	104.40000	106.20000	105.29683	0.50353

Forecast Span Summary									
Variable	Type	First Obs	Last Obs	NObs	NMiss	Min	Max	Mean	Standard Deviation
CPI	Dependent	JAN2012	SEP2017	69	0	104.40000	106.40000	105.38261	0.55676

Fixed Parameters in the Model		
Component	Parameter	Value
Slope	Error Variance	0

Preliminary Estimates of the Free Parameters		
Component	Parameter	Estimate
Irregular	Error Variance	2235.75255
Level	Error Variance	838.40721
Season	Error Variance	996.08045

Likelihood Based Fit Statistics	
Statistic	Value
Diffuse Log Likelihood	37.118
Diffuse Part of Log Likelihood	-13.93
Non-Missing Observations Used	63
Estimated Parameters	3
Initialized Diffuse State Elements	13
Normalized Residual Sum of Squares	50
AIC (smaller is better)	-68.24
BIC (smaller is better)	-62.5
AICC (smaller is better)	-67.71

Components: Irregular, Level, Slope (fixed), Seasonal (Trig, Length=12)

Section 2: Unobserved Components Model (UCM)

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Likelihood Based Fit Statistics	
Statistic	Value
HQIC (smaller is better)	-66.05
CAIC (smaller is better)	-59.5

Likelihood Optimization Algorithm Converged in 18 Iterations.

Final Estimates of the Free Parameters					
Component	Parameter	Estimate	Approx Std Error	t Value	Approx Pr > t
Irregular	Error Variance	5.94778E-10	6.58962E-7	0.00	0.9993
Level	Error Variance	0.00513	0.0010253	5.00	<.0001
Season	Error Variance	1.25548E-12	2.07609E-9	0.00	0.9995

Fit Statistics Based on Residuals	
Mean Squared Error	0.00852
Root Mean Squared Error	0.09228
Mean Absolute Percentage Error	0.07073
Maximum Percent Error	0.19139
R-Square	0.96825
Adjusted R-Square	0.96690
Random Walk R-Square	-0.71148
Amemiya's Adjusted R-Square	0.96420
Number of non-missing residuals used for computing the fit statistics = 50	

Significance Analysis of Components (Based on the Final State)			
Component	DF	Chi-Square	Pr > ChiSq
Irregular	1	0.00	0.9999
Level	1	1.177E7	<.0001
Slope	1	4.68	0.0305
Season	11	15.69	0.1530

Components: Irregular, Level, Slope (fixed), Seasonal (Trig, Length=12)

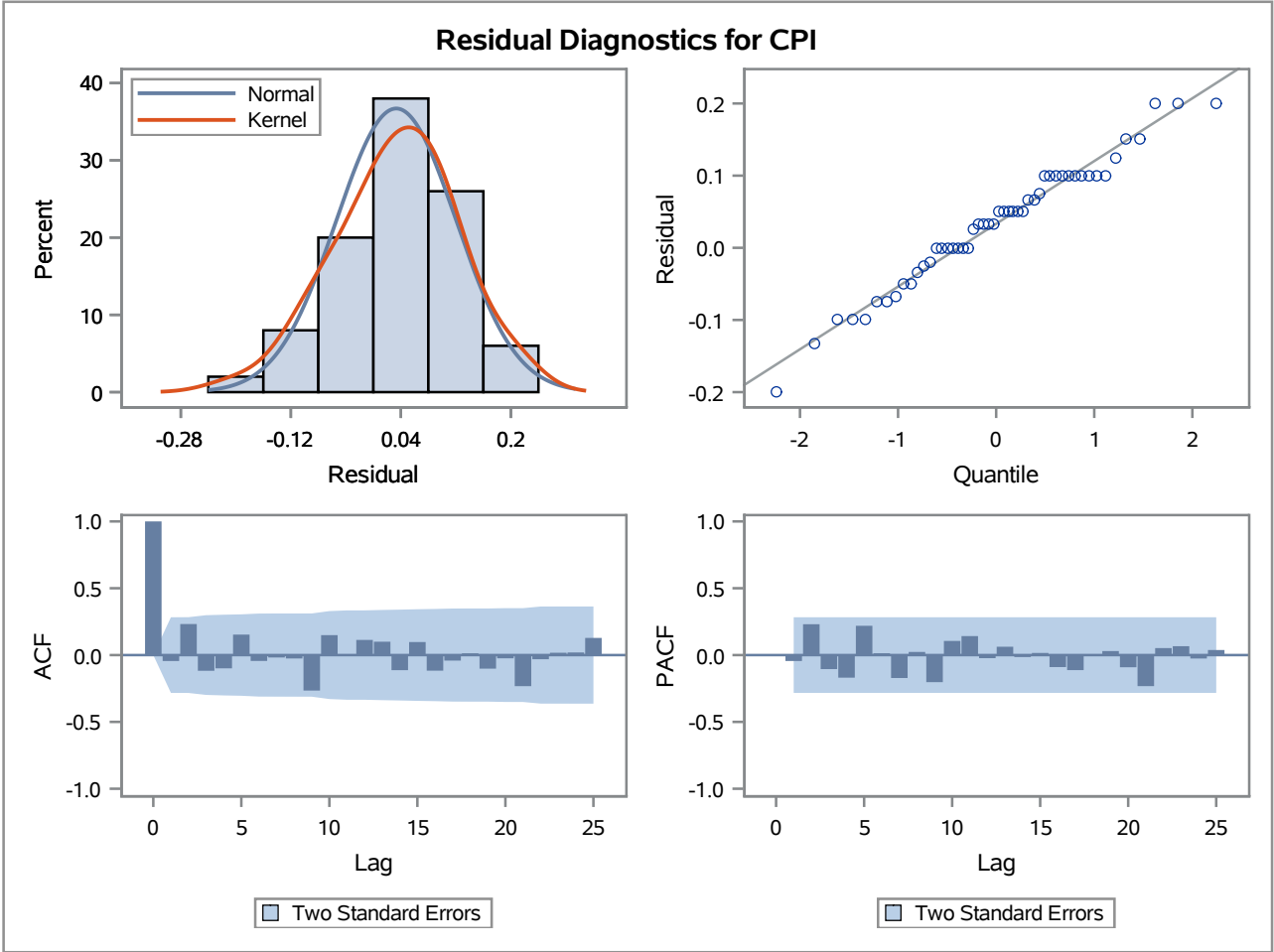
Section 2: Unobserved Components Model (UCM)

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Trend Information (Based on the Final State)		
Name	Estimate	Standard Error
Level	106.2031945	0.0309562
Slope	0.019722222	0.0091143

Summary of Seasons			
Name	Type	Season Length	Error Variance
Season	TRIG	12	1.25548E-12

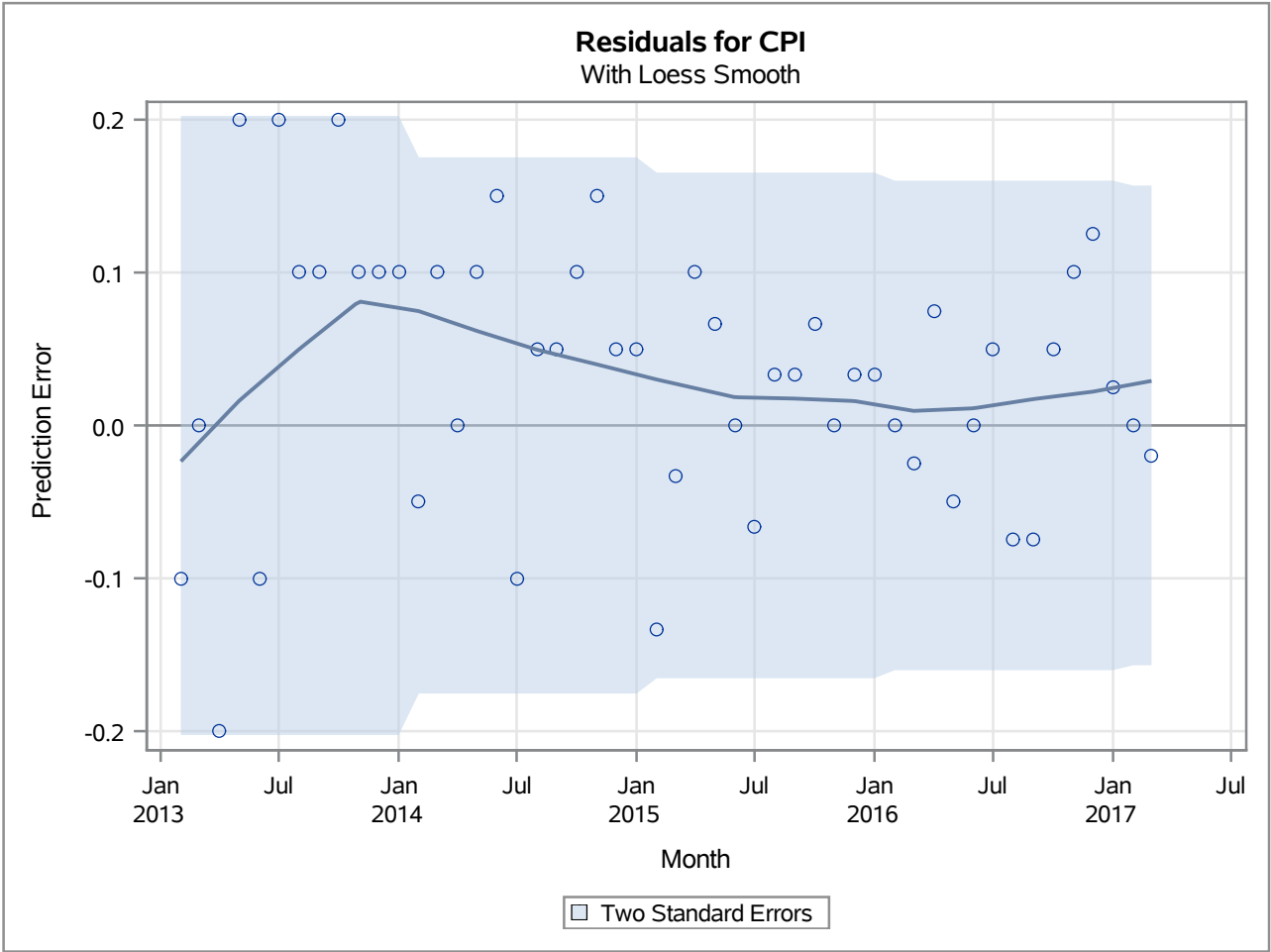
Outlier Summary							
Obs	Month	Break Type	Estimate	Standard Error	Chi-Square	DF	Pr > ChiSq
18	JUN2013	Additive Outlier	-0.15000	0.0566054	7.02	1	0.0081



Components: Irregular, Level, Slope (fixed), Seasonal (Trig, Length=12)

Section 2: Unobserved Components Model (UCM)

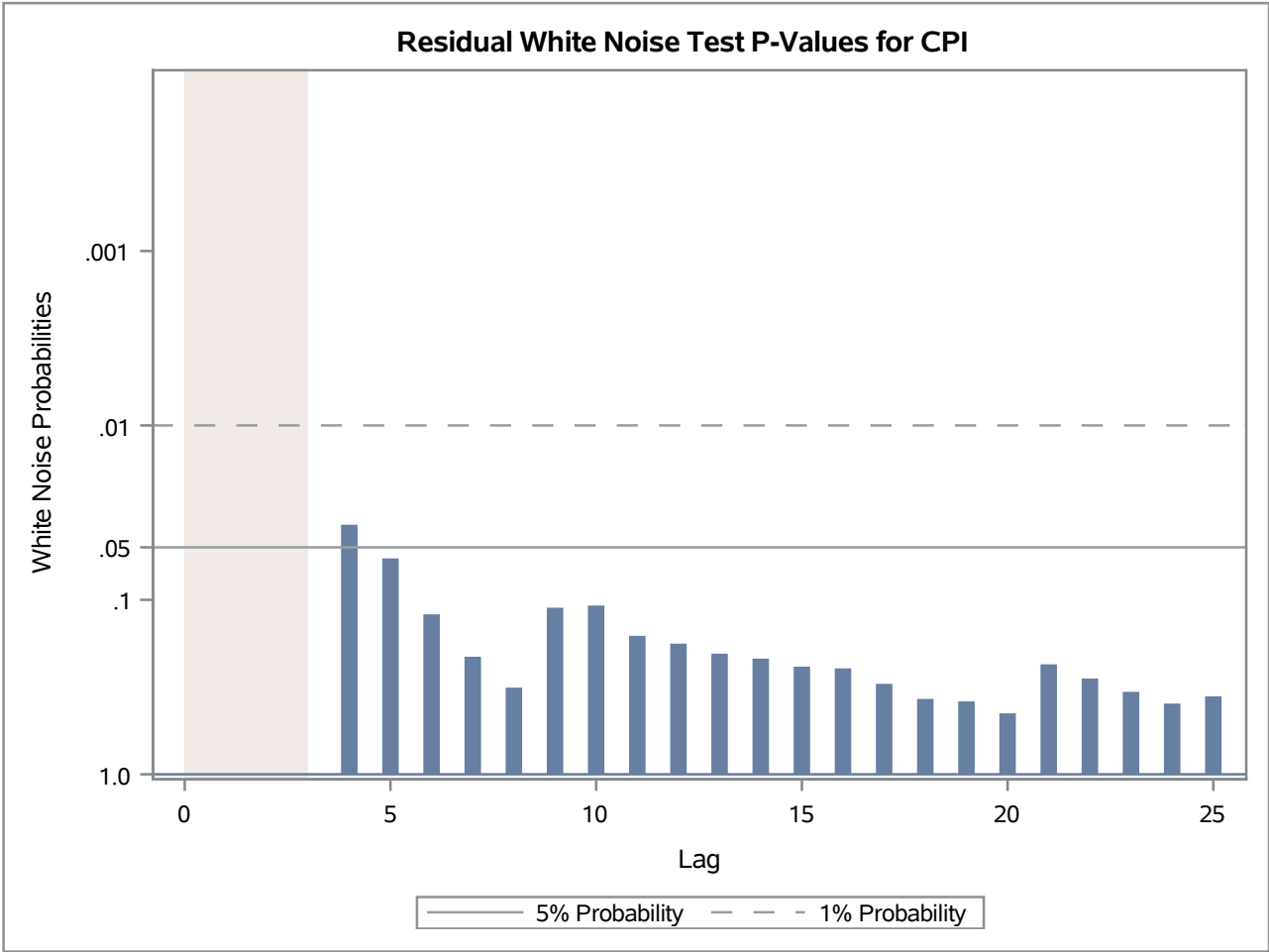
The UCM Procedure



Components: Irregular, Level, Slope (fixed), Seasonal (Trig, Length=12)

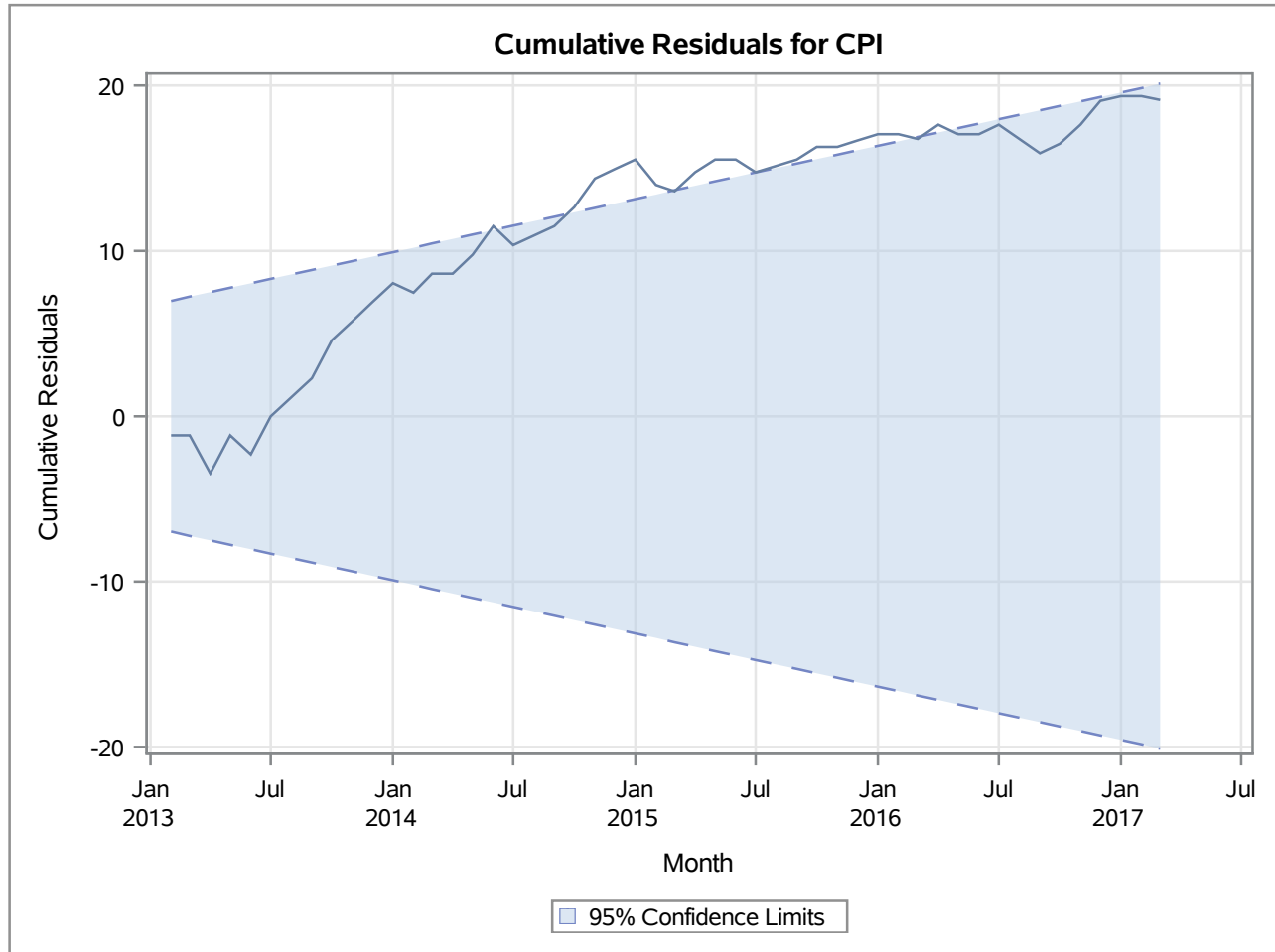
Section 2: Unobserved Components Model (UCM)

The UCM Procedure



Section 2: Unobserved Components Model (UCM)

The UCM Procedure



Forecasts for Variable CPI							
Obs	Month	Forecast	Standard Error	95% Confidence Limits		Actual	Residual
14	FEB2013	104.700000	0.10126	104.501536	104.898464	104.600000	-0.100000
15	MAR2013	104.600000	0.10126	104.401536	104.798464	104.600000	-1.27526E-8
16	APR2013	104.600000	0.10126	104.401536	104.798464	104.400000	-0.200000
17	MAY2013	104.300000	0.10126	104.101537	104.498464	104.500000	0.200000
18	JUN2013	104.500000	0.10126	104.301536	104.698463	104.400000	-0.100000
19	JUL2013	104.400000	0.10126	104.201536	104.598464	104.600000	0.200000
20	AUG2013	104.600000	0.10126	104.401536	104.798463	104.700000	0.100000
21	SEP2013	104.700000	0.10126	104.501536	104.898464	104.800000	0.100000
22	OCT2013	104.700000	0.10126	104.501536	104.898464	104.900000	0.200000
23	NOV2013	104.800000	0.10126	104.601536	104.998463	104.900000	0.100000
24	DEC2013	104.800000	0.10126	104.601536	104.998464	104.900000	0.100000
25	JAN2014	104.800000	0.10126	104.601536	104.998463	104.900000	0.100000
26	FEB2014	104.950000	0.08769	104.778126	105.121874	104.900000	-0.050000

Components: Irregular, Level, Slope (fixed), Seasonal (Trig, Length=12)

Section 2: Unobserved Components Model (UCM)

The UCM Procedure

Forecasts for Variable CPI							
Obs	Month	Forecast	Standard Error	95% Confidence Limits		Actual	Residual
27	MAR2014	104.900000	0.08769	104.728126	105.071874	105.000000	0.100000
28	APR2014	104.900000	0.08769	104.728126	105.071874	104.900000	1.74913E-8
29	MAY2014	104.900000	0.08769	104.728126	105.071874	105.000000	0.100000
30	JUN2014	104.950000	0.08769	104.778126	105.121874	105.100000	0.150000
31	JUL2014	105.200000	0.08769	105.028126	105.371874	105.100000	-0.100000
32	AUG2014	105.150000	0.08769	104.978126	105.321874	105.200000	0.050000
33	SEP2014	105.250000	0.08769	105.078126	105.421874	105.300000	0.050000
34	OCT2014	105.300000	0.08769	105.128126	105.471874	105.400000	0.100000
35	NOV2014	105.350000	0.08769	105.178126	105.521874	105.500000	0.150000
36	DEC2014	105.450000	0.08769	105.278126	105.621874	105.500000	0.050000
37	JAN2015	105.450000	0.08769	105.278126	105.621874	105.500000	0.050000
38	FEB2015	105.533333	0.08268	105.371289	105.695378	105.400000	-0.133333
39	MAR2015	105.433333	0.08268	105.271289	105.595378	105.400000	-0.033333
40	APR2015	105.300000	0.08268	105.137955	105.462045	105.400000	0.100000
41	MAY2015	105.433333	0.08268	105.271289	105.595378	105.500000	0.066667
42	JUN2015	105.500000	0.08268	105.337955	105.662045	105.500000	9.048634E-9
43	JUL2015	105.566667	0.08268	105.404622	105.728711	105.500000	-0.066667
44	AUG2015	105.566667	0.08268	105.404622	105.728711	105.600000	0.033333
45	SEP2015	105.666667	0.08268	105.504622	105.828711	105.700000	0.033333
46	OCT2015	105.733333	0.08268	105.571289	105.895378	105.800000	0.066667
47	NOV2015	105.800000	0.08268	105.637955	105.962045	105.800000	8.044651E-9
48	DEC2015	105.766667	0.08268	105.604622	105.928711	105.800000	0.033333
49	JAN2016	105.766667	0.08268	105.604622	105.928711	105.800000	0.033333
50	FEB2016	105.800000	0.08005	105.643101	105.956899	105.800000	1.114981E-8
51	MAR2016	105.825000	0.08005	105.668101	105.981899	105.800000	-0.025000
52	APR2016	105.725000	0.08005	105.568101	105.881899	105.800000	0.075000
53	MAY2016	105.850000	0.08005	105.693101	106.006899	105.800000	-0.050000
54	JUN2016	105.800000	0.08005	105.643101	105.956899	105.800000	-9.15588E-9
55	JUL2016	105.850000	0.08005	105.693101	106.006899	105.900000	0.050000
56	AUG2016	105.975000	0.08005	105.818101	106.131899	105.900000	-0.075000
57	SEP2016	105.975000	0.08005	105.818101	106.131899	105.900000	-0.075000
58	OCT2016	105.950000	0.08005	105.793101	106.106899	106.000000	0.050000
59	NOV2016	106.000000	0.08005	105.843101	106.156899	106.100000	0.100000

Components: Irregular, Level, Slope (fixed), Seasonal (Trig, Length=12)

Section 2: Unobserved Components Model (UCM)**The UCM Procedure**

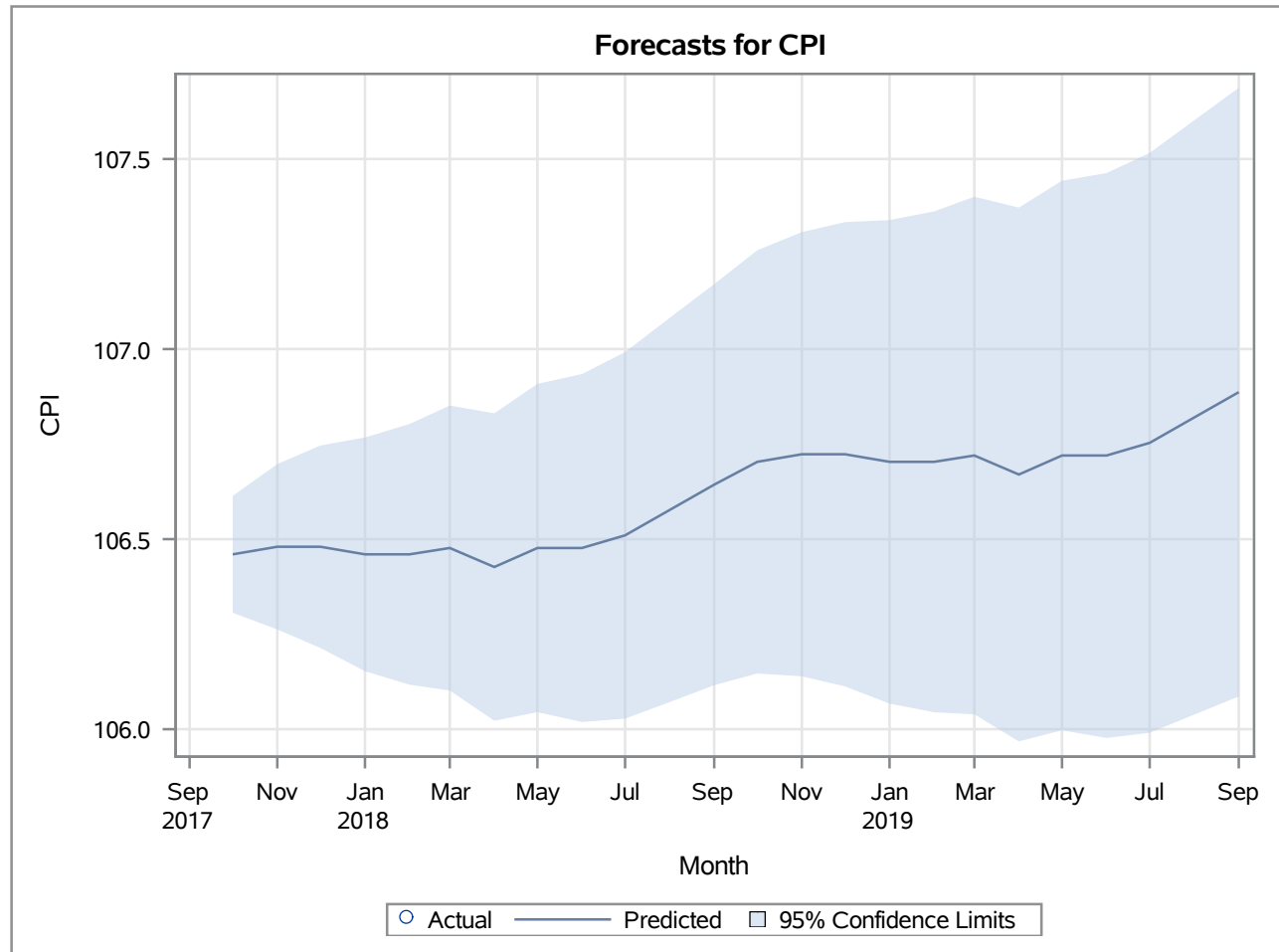
Forecasts for Variable CPI							
Obs	Month	Forecast	Standard Error	95% Confidence Limits		Actual	Residual
60	DEC2016	106.075000	0.08005	105.918101	106.231899	106.200000	0.125000
61	JAN2017	106.175000	0.08005	106.018101	106.331899	106.200000	0.025000
62	FEB2017	106.200000	0.07843	106.046271	106.353729	106.200000	7.799784E-9
63	MAR2017	106.220000	0.07843	106.066271	106.373729	106.200000	-0.020000
64	APR2017	106.140000	0.07843	105.986271	106.293729	106.200000	0.060000
65	MAY2017	106.240000	0.07843	106.086271	106.393729	106.300000	0.060000
66	JUN2017	106.300000	0.07843	106.146271	106.453729	106.300000	7.029541E-9
67	JUL2017	106.360000	0.07843	106.206271	106.513729	106.200000	-0.160000
68	AUG2017	106.260000	0.07843	106.106271	106.413729	106.300000	0.040000
69	SEP2017	106.360000	0.07843	106.206271	106.513729	106.400000	0.040000
70	OCT2017	106.460000	0.07843	106.306271	106.613729	.	.
71	NOV2017	106.480000	0.11092	106.262594	106.697406	.	.
72	DEC2017	106.480000	0.13585	106.213733	106.746267	.	.
73	JAN2018	106.460000	0.15687	106.152542	106.767458	.	.
74	FEB2018	106.460000	0.17490	106.117207	106.802793	.	.
75	MAR2018	106.476667	0.19123	106.101856	106.851477	.	.
76	APR2018	106.426667	0.20628	106.022366	106.830968	.	.
77	MAY2018	106.476667	0.22030	106.044885	106.908448	.	.
78	JUN2018	106.476667	0.23348	106.019052	106.934282	.	.
79	JUL2018	106.510000	0.24596	106.027934	106.992066	.	.
80	AUG2018	106.576667	0.25783	106.071331	107.082002	.	.
81	SEP2018	106.643333	0.26918	106.115754	107.170913	.	.
82	OCT2018	106.703333	0.28401	106.146691	107.259975	.	.
83	NOV2018	106.723333	0.29810	106.139073	107.307594	.	.
84	DEC2018	106.723333	0.31155	106.112702	107.333965	.	.
85	JAN2019	106.703333	0.32445	106.067424	107.339243	.	.
86	FEB2019	106.703333	0.33609	106.044606	107.362061	.	.
87	MAR2019	106.720000	0.34734	106.039219	107.400781	.	.
88	APR2019	106.670000	0.35824	105.967858	107.372142	.	.
89	MAY2019	106.720000	0.36882	105.997128	107.442872	.	.
90	JUN2019	106.720000	0.37910	105.976976	107.463024	.	.
91	JUL2019	106.753333	0.38911	105.990689	107.515977	.	.

Components: Irregular, Level, Slope (fixed), Seasonal (Trig, Length=12)

Section 2: Unobserved Components Model (UCM)

The UCM Procedure

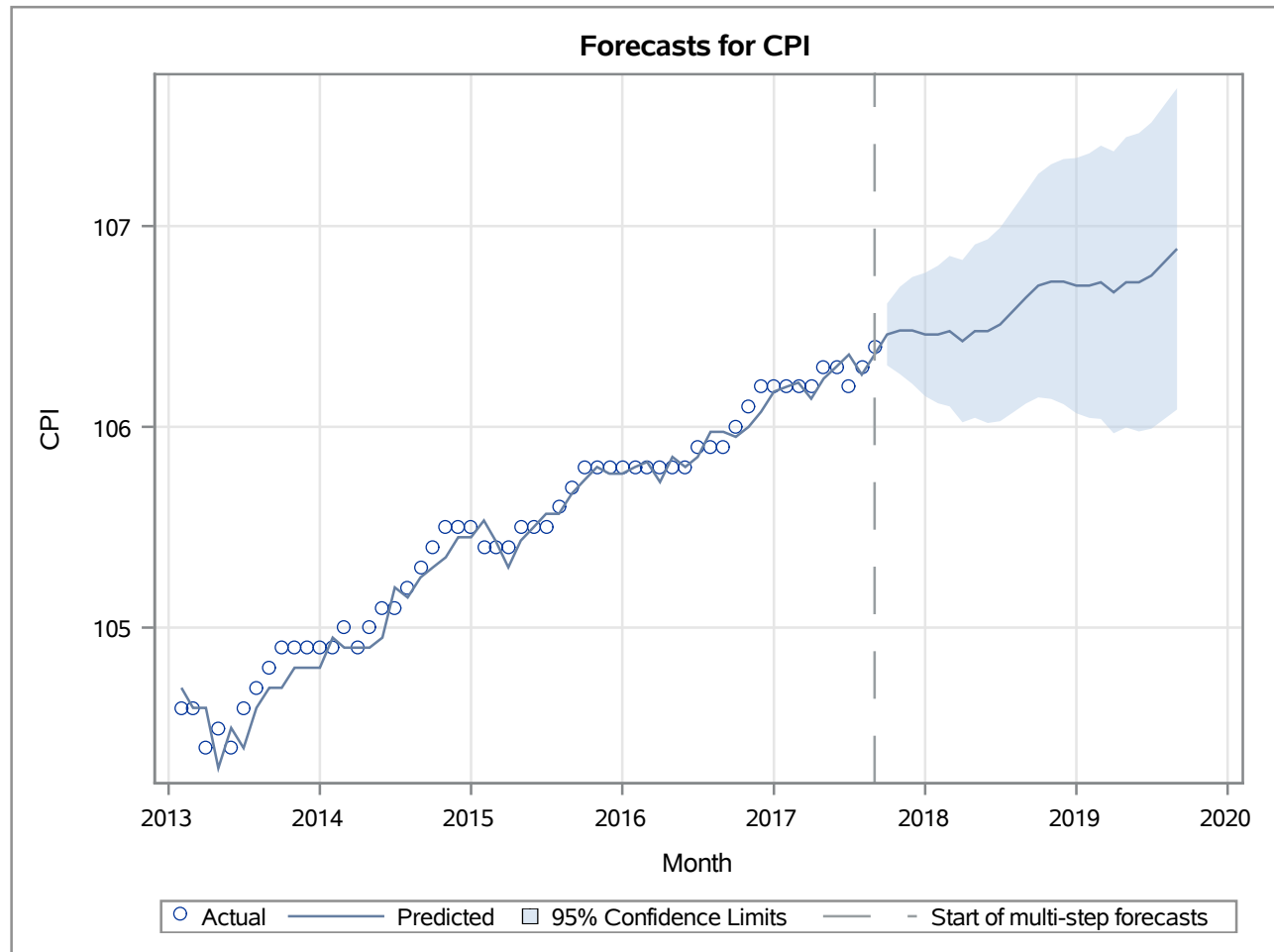
Forecasts for Variable CPI							
Obs	Month	Forecast	Standard Error	95% Confidence Limits		Actual	Residual
92	AUG2019	106.820000	0.39887	106.038228	107.601772	.	.
93	SEP2019	106.886667	0.40840	106.086225	107.687109	.	.



Components: Irregular, Level, Slope (fixed), Seasonal (Trig, Length=12)

Section 2: Unobserved Components Model (UCM)

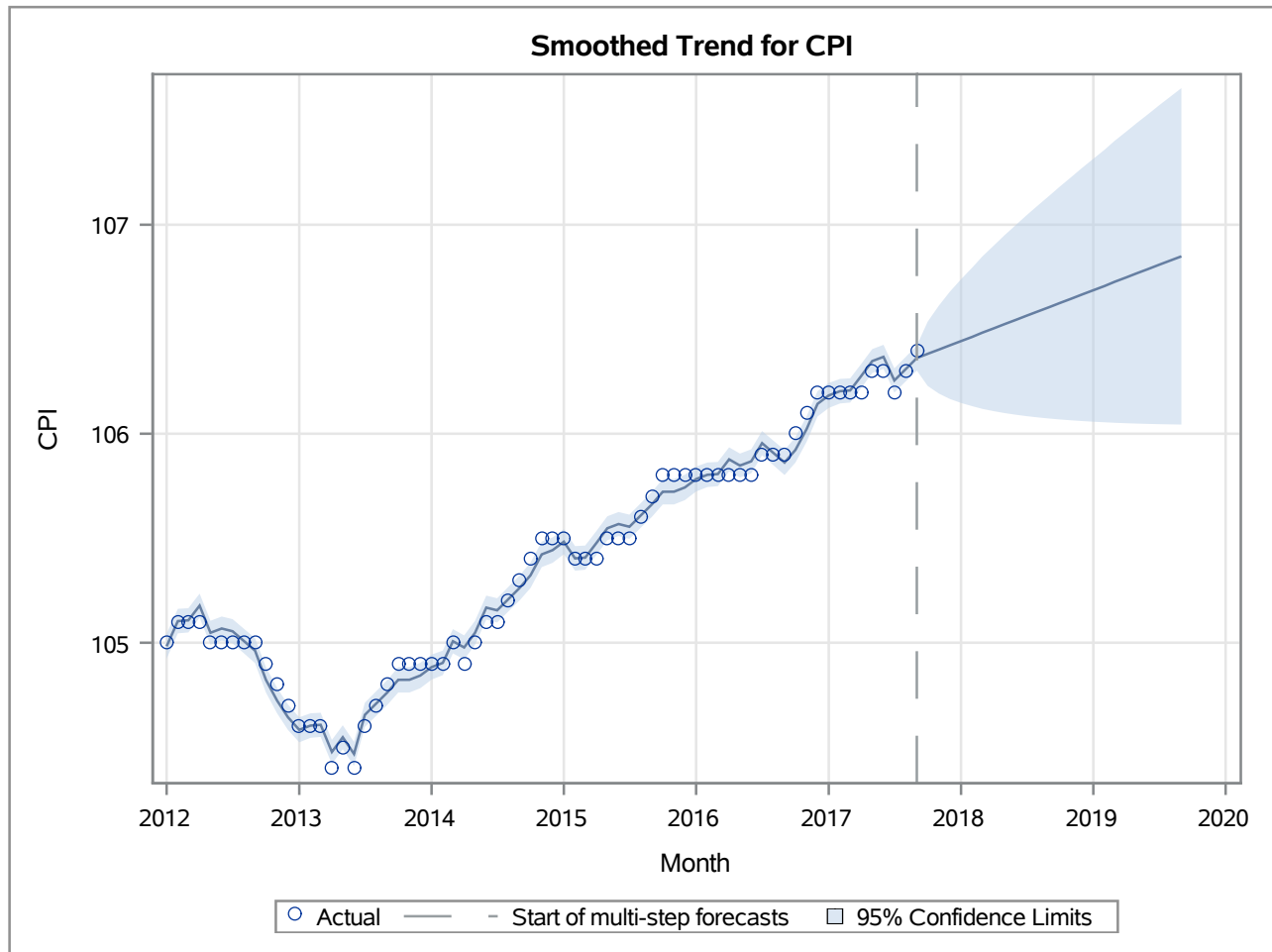
The UCM Procedure



Components: Irregular, Level, Slope (fixed), Seasonal (Trig, Length=12)

Section 2: Unobserved Components Model (UCM)

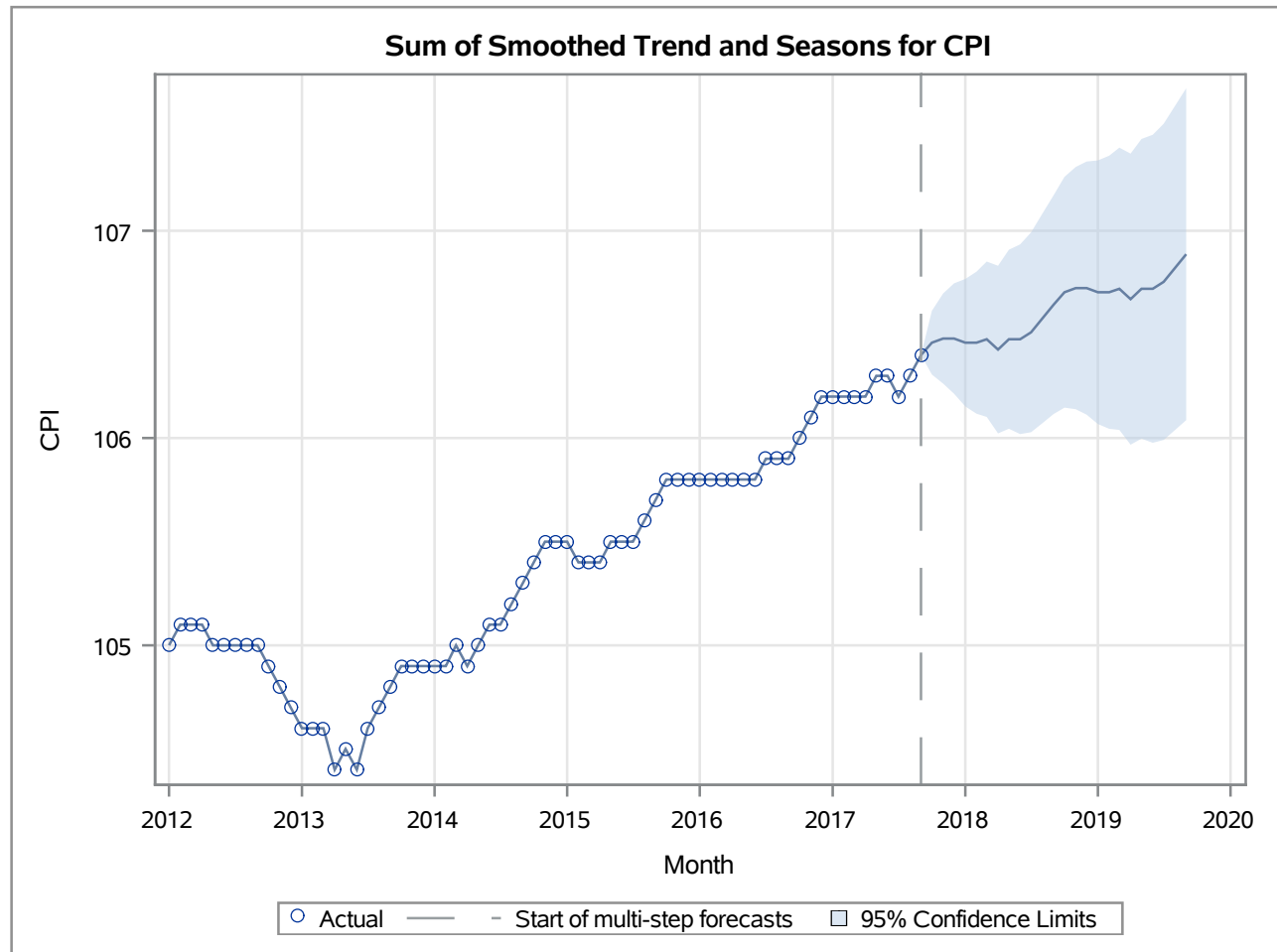
The UCM Procedure



Components: Irregular, Level, Slope (fixed), Seasonal (Trig, Length=12)

Section 2: Unobserved Components Model (UCM)

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Components: Irregular, Level, Slope (fixed), Seasonal (Trig, Length=12)

Section 3: Forecast Comparison — UCM vs Regression Models

Month	Forward_Selection	Backward_Selection	MaxR	UCM	Observed
Oct-2017	106.408	106.408	106.358	106.460	106.4
Nov-2017	106.408	106.408	106.359	106.480	106.5
Dec-2017	106.341	106.341	106.292	106.480	106.6
Jan-2018	106.272	106.272	106.203	106.460	106.6
Feb-2018	106.205	106.205	106.127	106.460	106.6
Mar-2018	106.205	106.205	106.120	106.477	106.7

UCM forecasts are directionally closer to observed CPI values